1. **This example is adapted from a real production application, with details modified for confidentiality.** 1 point

You are a renowned researcher in the City of Peacetopia. The residents of Peacetopia share a unique characteristic: they are afraid of birds. To protect them, you are tasked with developing an algorithm that will detect any bird flying over Peacetopia and alert the population.

The City Council provides you with a dataset of 10,000,000 images of the sky above Peacetopia, captured by the city's security cameras. They are labeled:

- **y = 0: There is no bird on the image**
- **y = 1: There is a bird on the image**

Your goal is to create an algorithm capable of classifying new images taken by security cameras in Peacetopia. You have several decisions to make regarding the evaluation metric and how to structure your data into train/dev/test sets.

The City Council specifies that they want an algorithm that:

1. **Has high accuracy.**
2. **Operates quickly and takes only a short time to classify a new image.**
3. **Requires minimal memory, allowing it to run on a small processor attached to various security cameras.**

True or False: You discuss with them the need for a singular evaluation metric to guide development.

- ( ) False
- (•) True

2. **The city asks for your help in further defining the criteria for accuracy, runtime, and memory. How would you suggest they identify the criteria?** 1 point

- ( ) Suggest to them that they focus on whichever criterion is to be optimized and then eliminate the other two.
- ( ) Suggest that they purchase more infrastructure to ensure the model runs quickly and accurately.
- (•) Suggest to them that they define which criterion is to be optimized. Then, set thresholds for the other two.

3. **True or False: The essential difference between an optimizing metric and satisficing metrics is the priority assigned by the stakeholders.** 1 point

- (•) True
- ( ) False

4. **Before implementing your algorithm, you need to split your data into train/dev/test sets. You have a dataset of 10,000,000 examples. Which of these do you think is the best choice?** 1 point

( )

Train	Dev	Test
6,000,000	3,000,000	1,000,000

( )

Train	Dev	Test
6,000,000	1,000,000	3,000,000

( )

Train	Dev	Test
3,333,334	3,333,333	3,333,333

(•)

Train	Dev	Test
9,500,000	250,000	250,000

5. **After setting up your train/dev/test sets, the City Council comes across another 1,000,000 images, called the "citizens' data." Apparently, the citizens of Peacetopia are so scared of birds that they volunteered to take pictures of the sky and label them, thus contributing these additional 1,000,000 images. These images have a different distribution from the images the City Council originally provided, but you think they could help your algorithm. Notice that adding this additional data to the training set will make the distribution of the training set different from the distribution of the dev and test sets.** 1 point

True or False: You should not add the citizens' data to the training set, because if the training distribution is different from the dev and test sets, then this will not allow the model to perform well on the test set.

- ( ) True
- (•) False

6. **One member of the City Council wants to add 1,000,000 citizen data images to the development (dev) set. Your original data is from security cameras, and you object to adding the citizen data because: (Choose all that apply)** 1 point

- [ ] The 1,000,000 citizen data images do not have a consistent input-output relationship as the security camera data.
- [x] This would cause the dev and test set distributions to become different. This is a bad idea because you're not aiming where you want to hit.
- [ ] A bigger test set will slow down the speed of iterating because of the computational expense of evaluating models on the test set.
- [x] The dev set no longer reflects the distribution of data (security cameras) you most care about.

7. **You train a system, and the train/dev set errors are 3.5% and 4.0% respectively. You decide to try regularization to close the train/dev error gap. Do you agree?** 1 point

- (•) No, because you do not know what the human performance level is.
- ( ) Yes, because this shows your bias is higher than your variance.
- ( ) Yes, because having a 4.0% training error shows you have a high bias.
- ( ) No, because this shows your variance is higher than your bias.

8. **You ask a few people to label a bird species dataset to determine human-level performance. The following error rates were recorded:** 1 point

Bird watching expert #1	0.3% error
Bird watching expert #2	0.5% error
Normal person #1 (not a bird watching expert)	1.0% error
Normal person #2 (not a bird watching expert)	1.2% error

If your goal is to use "human-level performance" as an estimate for Bayes error, how would you define "human-level performance" in this scenario?

- (x) 0.3% (The lowest error rate achieved by an expert)
- ( ) 0.4% (Average of the two experts' error rates)
- ( ) 0.0% (Perfect accuracy, representing an unattainable ideal)
- ( ) 0.75% (Average of all four error rates)

9. Which of the below shows the optimal order of accuracy from worst to best? 1 point

- ( ) Human-level performance -> the learning algorithm's performance -> Bayes error.
- (x) The learning algorithm's performance -> human-level performance -> Bayes error.
- ( ) Human-level performance -> Bayes error -> the learning algorithm's performance.
- ( ) The learning algorithm's performance -> Bayes error -> human-level performance.

10. Which of the following best describes the most effective next step in your project, given the following performance metrics? 1 point

- **Human-level performance: 0.1%**
- **Training set error: 2.0%**
- **Dev set error: 2.1%**

- ( ) Evaluate the test set to determine the variance.
- ( ) Deploy the model to target devices to evaluate against satisficing metrics.
- ( ) Continue tuning until the training set error matches human-level performance, focusing solely on the optimizing metric.
- (x) Prioritize actions to decrease bias by increasing model complexity, as the training error significantly exceeds human-level performance.

11. You also evaluate your model on the test set and find the following: 1 point

Human-level performance	0.1%
Training set error	2.0%
Dev set error	2.1%
Test set error	7.0%

What does this mean? (Check the two best options.)

- [x] You should try to get a bigger dev set.
- [x] You have overfit to the dev set.
- [ ] You have underfitted to the dev set.
- [ ] You should get a bigger test set.

12. After working on this project for a year, you finally achieve: Human-level performance, 0.10%, Training set error, 0.05%, Dev set error, 0.05%. Which of the followingare true? (Check all that apply.) 1 point

- [ ] It is highly unlikely this result is purely a statistical anomaly, but statistical noise may still contribute to the error.
- [x] You are close to Bayes error and there is a possibility of overfitting.
- [ ] With only 0.05% further progress to make, you should quickly be able to close the remaining gap to 0%.
- [x] All or almost all of the avoidable bias has been accounted for.

13. Your system is now very accurate but has a higher false negative rate than the City Council of Peacetopia would like. What is your best next step? 1 point

- ( ) Expand your model size to account for more corner cases.
- ( ) Pick false negative rate as the new metric, and use this new metric to drive all further development.
- ( ) Look at all the models you've developed during the development process and find the one with the lowest false negative error rate.
- (x) Reset your "target" (metric) for the team and tune to it.

14. Over the last few months, a new species of bird has been slowly migrating into the area, so the performance of your system slowly degrades because your data is being tested on a new type of data. There are only 1,000 images of the new species. The city expects a better system from you within the next 3 months. 1 point

Which of these should you do first?

- ( ) Add pooling layers to downsample features to accommodate the new species.
- ( ) Split them between dev and test and re-tune.
- (x) Augment your data to increase the number of images of the new bird species.
- ( ) Put the new species' images in training data to learn their features.

15. The City Council thinks that having more cats in the city would help scare off birds. They are so happy with your work on the Bird detector that they also hire you to build a Cat detector. 1 point

You have a huge dataset of 100,000,000 cat images. Training on this data takes about two weeks.

Which of the statements do you agree with? (Check all that agree.)

- [ ] Reducing the model complexity will allow the use of the larger dataset but preserve accuracy.
- [x] Lowering the number of images will reduce training time and likely allow for an acceptable tradeoff between iteration speed and accuracy.
- [x] This significantly impacts iteration speed.